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Principles of Data Science II
30 April, 2025

Capstone Project: Assessing Professor Effectiveness

Preprocessing:

The datasets rmpCapstoneNum.csv and rmpCapstoneQual.csv were loaded without headers, as the first row directly contained data, so appropriate column names were manually assigned. And then I merged the two datasets. After that, rows with missing values in the Number of Ratings field were dropped, under the assumption that missing number of ratings indicates an invalid record. To minimize the impact of extreme values based on very few ratings, only professors with at least 5 ratings were retained for analysis. Primary data operations and calculations were conducted using pandas and numpy libraries. Before conducting any statistical test or model training, data cleaning was performed to ensure that only complete and valid entries were used in all analyses.

Random Number Generator (RNG) Seeding:

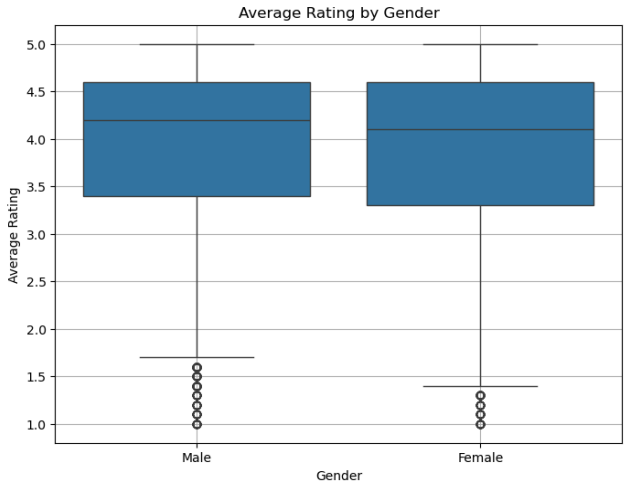
To ensure reproducibility and protect against plagiarism, the random number generator was seeded using my N-number (N13160240), with the prefix letter removed.

Q1:

I excluded entries where both the 'Male' and 'Female' indicators were 0. Then, I compared the average ratings of male and female professors using the Mann-Whitney U test, a non-parametric method that does not assume normality.

The median rating for male professors was 4.2, compared to 4.1 for female professors. The U-statistic is 50,901,762.5 and the p-value is 0.00084<0.005, indicating a statistically significant difference. This suggests that male professors tend to receive slightly higher ratings than female professors.

As shown in the boxplot, male professors exhibit a slightly higher median average rating compared to female professors. Although the difference is visually subtle, the statistical analysis confirms that it is highly significant due to the large sample size.

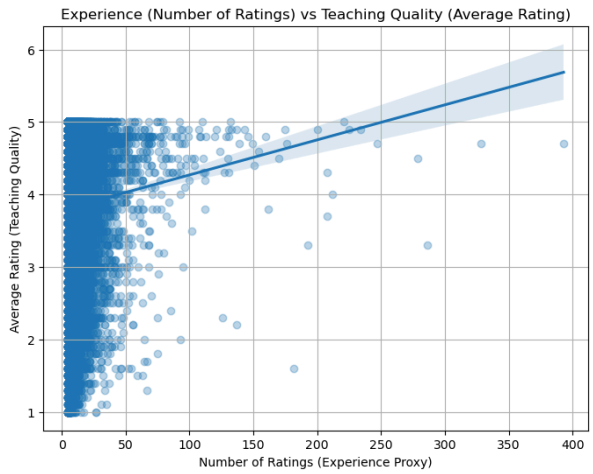


Q2:

I conducted a Pearson correlation test between Number of Ratings (as a proxy for experience) and Average Rating (as a proxy for teaching quality). The threshold for statistical significance was set at $\alpha = 0.005$.

The Pearson correlation coefficient was $r = 0.058$ with a p-value of $p < 0.00001$. Since the p-value is far less than 0.005, we conclude that there is a statistically significant relationship between experience and teaching quality.

However, the correlation coefficient is very close to zero, indicating that although statistically significant, the relationship is extremely weak and may not be practically meaningful. And there is an effect of experience on the quality of teaching.

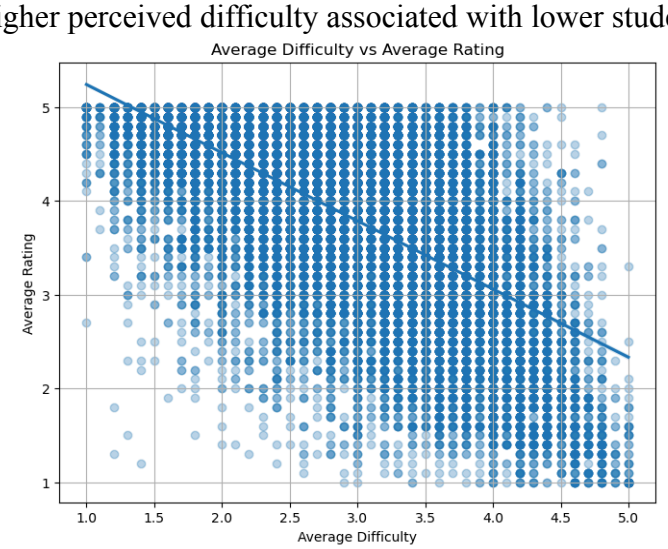


Q3:

I conducted a Pearson correlation test between Average Difficulty and Average Rating. The threshold for statistical significance was set at $\alpha = 0.005$.

The Pearson correlation coefficient was $r = -0.619$ with a p-value of $p < 0.00001$. Since the p-value is far less than 0.005, we conclude that there is a statistically significant relationship between average difficulty and average rating. Specifically, the negative correlation indicates that professors perceived as more difficult tend to receive lower average ratings.

The scatter plot also shows a clear negative relationship between average difficulty and average rating, with higher perceived difficulty associated with lower student ratings.

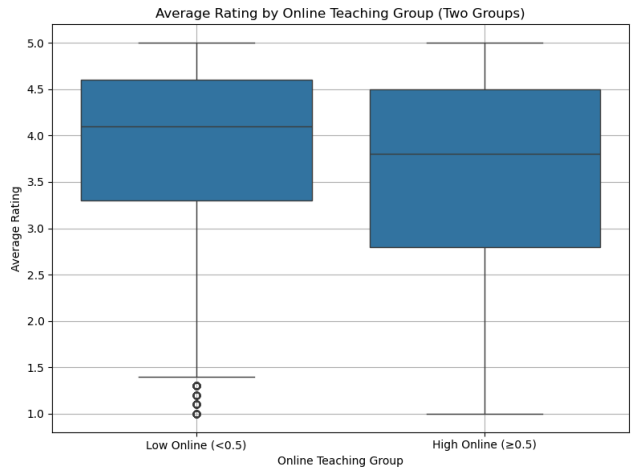


Q4:

I calculated the proportion of online ratings to total ratings for each professor and divided the data into two groups: Low Online (<0.5), High Online (≥0.5)

To compare student ratings between these groups, I used a Mann-Whitney U test, a non-parametric test that does not assume normality. The median rating for Low Online professors was 4.1, compared to 3.8 for High Online professors. The U-statistic was 14,115,560.0, and the p-value was 2.71e-11<0.005, indicating a statistically significant difference between the two groups. Professors who teach more than half of their classes online tend to receive lower student ratings.

The boxplot below illustrates this difference in rating distributions between the two groups.



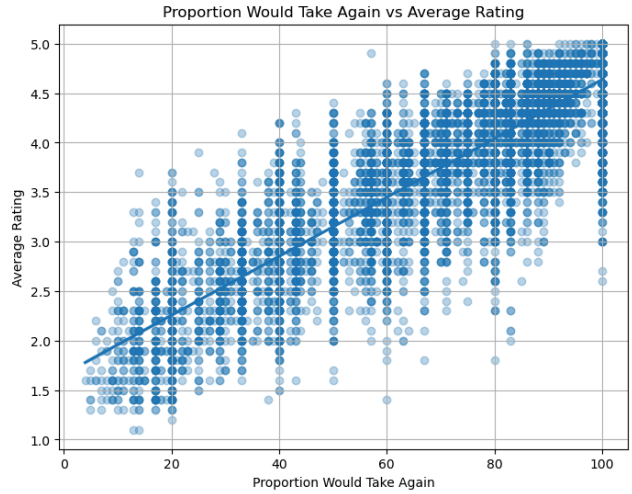
Q5:

Before conducting the analysis, I filtered the dataset to include only entries with non-missing values in both Proportion Would Take Again and Average Rating.

I then conducted a Pearson correlation test between Proportion Would Take Again and Average Rating. The significance threshold was set at $\alpha = 0.005$.

The Pearson correlation coefficient was $r = 0.880$ with a p-value of $p < 0.00001$. Since the p-value is far less than 0.005, we conclude that there is a statistically significant relationship between the proportion of students willing to retake a professor's class and the professor's average rating. The strong positive correlation indicates that professors with higher willingness-to-retake proportions tend to have higher average ratings.

The scatter plot shows a strong positive relationship between the proportion of students willing to retake a professor's class and the professor's average rating.

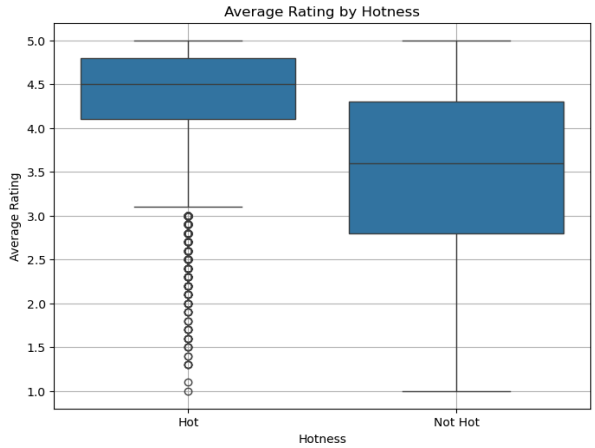


Q6:

To examine whether receiving a “Hot Pepper” is associated with higher ratings, I compared the average ratings of professors who received a pepper versus those who did not. I used a Mann-Whitney U test to assess statistical significance, and reported the median of each group for a more robust comparison.

The median rating for "Hot" professors was 4.5, while "Not Hot" professors had a median of 3.6. The Mann-Whitney U test yielded a U-statistic of 121,910,581.5 and a p-value < 0.00001, indicating a highly significant difference. This suggests that professors perceived as "hot" tend to receive substantially higher student ratings.

The boxplot clearly shows that hot professors have higher median and overall average ratings compared to non-hot professors.

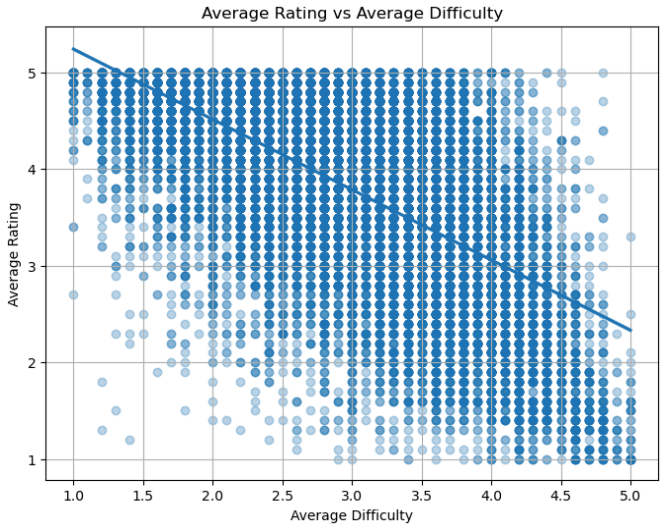


Q7:

I constructed a simple linear regression model using Average Difficulty as the predictor and Average Rating as the response variable. Before fitting the model, I filtered the data to exclude entries with missing values in either field.

The regression model yielded an R^2 of 0.3832 and an RMSE of 0.7436. The R^2 value suggests that average difficulty explains about 38% of the variation in average ratings, indicating a moderate relationship. The RMSE indicates that the average prediction error is approximately 0.74 points on a 5-point scale.

The scatter plot shows a moderate negative relationship between average difficulty and average rating, with higher difficulty generally associated with lower ratings. The data points are fairly dispersed around the regression line.

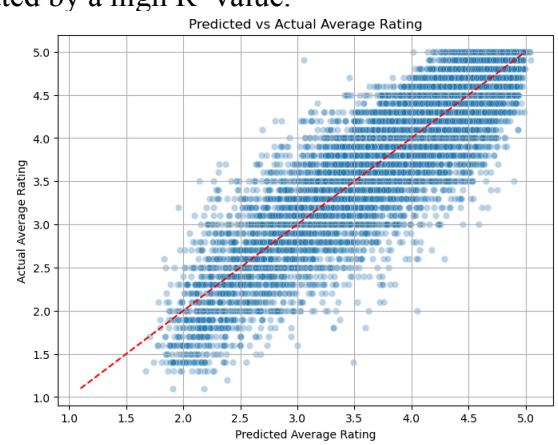


Q8:

I constructed a multiple linear regression model using all available factors to predict Average Rating. Before fitting the model, I filtered the dataset to exclude entries with missing values. I also assessed multicollinearity among predictors using the Variance Inflation Factor (VIF).

The regression model yielded an R^2 of 0.8101 and an RMSE of 0.3687. The regression model yielded an R^2 of 0.8101 and an RMSE of 0.3687. Significant predictors included Average Difficulty (negative effect), Received a Pepper (positive effect), Proportion Would Take Again (positive effect), Male, and Female. Number of Ratings and Number of Online Ratings were not statistically significant. VIF values for all predictors were low (all below 2), indicating no major multicollinearity concerns.

The scatter plot below shows the predicted versus actual average ratings. The points closely cluster around the diagonal line, supporting the model's strong predictive performance as indicated by a high R^2 value.



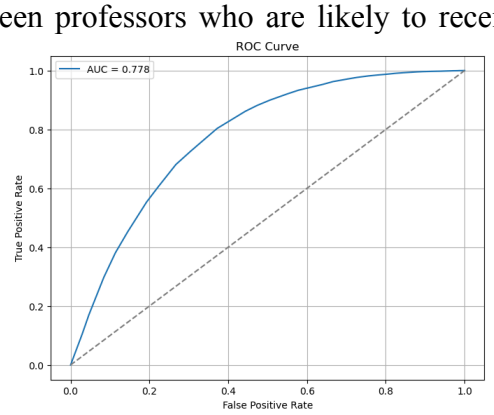
Feature	Beta Coefficient	p-value	Statistically Significant (p < 0.005)?	Interpretation
Average Difficulty	-0.1994	0.0000	✅	Higher difficulty is associated with lower ratings.
Number of Ratings	-0.0002	0.3609	❌	Not statistically significant.
Received a Pepper	+0.2049	0.0000	✅	Having a Hot Pepper is associated with higher ratings.
Proportion Would Take Again	+0.0249	0.0000	✅	Higher proportion willing to retake is associated with higher ratings.
Number of Online Ratings	-0.0005	0.7659	❌	Not statistically significant.
Male	+0.0508	0.0000	✅	Male professors tend to have slightly higher ratings.
Female	+0.0269	0.0004	✅	Female professors also tend to have slightly higher ratings.

Q9:

I built a logistic regression model using Average Rating as the sole predictor to classify whether a professor received a "Hot Pepper." Before training the model, I filtered out missing data and ensured that all ratings were based on at least 5 reviews to maintain data quality.

The model achieved an accuracy of 0.7110 and an AUC of 0.7776. The ROC curve below illustrates the model's performance, showing a reasonable trade-off between the true positive rate and the false positive rate.

Since the number of professors who received a "Hot Pepper" is much smaller than those who did not, there is a class imbalance issue. However, using AUC as a quality metric helps account for this imbalance because AUC is insensitive to the proportion of positive and negative classes. The AUC value of 0.7776 and the ROC curve below indicate that the model can effectively distinguish between professors who are likely to receive a "Hot Pepper" and those who are not.

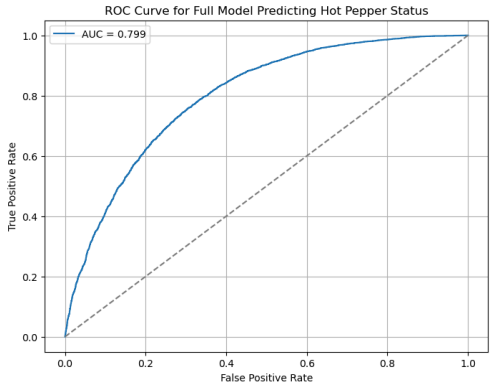


Q10:

I built a logistic regression model using all available factors to classify whether a professor received a "Hot Pepper." Before training the model, I filtered out missing data and ensured all ratings were based on at least 5 reviews.

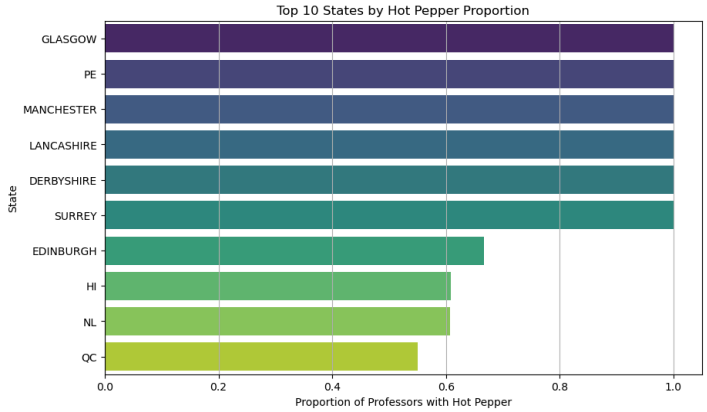
The model achieved an accuracy of 0.7240 and an AUC of 0.7991. The ROC curve below shows the model's performance. Compared to the previous model based only on Average Rating (Accuracy = 0.7110, AUC = 0.7776), the full model shows a slight improvement in both accuracy and AUC. This suggests that incorporating additional factors beyond average rating helps better predict whether a professor receives a "Hot Pepper."

As with the previous model, there remains a noticeable class imbalance. However, using AUC as a quality metric mitigates this concern because AUC evaluates ranking performance rather than relying on absolute prediction counts. The ROC curve below demonstrates the model's classification performance. The curve bends significantly toward the top-left corner, indicating a high true positive rate and a low false positive rate, with an AUC of 0.7991.



Question for Extra Credit:

I analyzed the proportion of professors who received a "Hot Pepper" across different states. After merging the numerical and qualitative datasets and filtering out entries with missing data or insufficient ratings (fewer than 5), I calculated the proportion of Hot Pepper recipients for each state by taking the mean of the Received a Pepper indicator within each state.

This suggests meaningful regional variation in how students perceive and reward professors with the "Hot Pepper" label. For example, locations such as Glasgow, Manchester, and Surrey, which appear at the top of the list, may reflect cultural norms that place greater emphasis on charisma, approachability, or personal appearance in academic settings. Such differences could stem from local campus culture, varying expectations of professionalism, or even differences in how seriously students treat RateMyProfessors as a platform.

The bar chart below shows the top 10 states ranked by the proportion of professors who received a "Hot Pepper."